

Tankut Can

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Research Interests

Science of artificial intelligence, statistical physics of neural systems, theoretical neuroscience and dynamical systems, memory and cognition in humans and machines

Current Projects

- **Science of Artificial Intelligence** : Building theory for different facets of language modeling. This includes a renormalization group analysis of tokenization compression algorithms, and exploring symmetries of tokenized large language models.
- **Statistical Mechanics of Human Memory**: Understand human memory on narrative recall tasks, with a combination of experiment and theory. Involves the use of AI for quantitative analysis of language data. Also includes developing statistical mechanical models to explain experimental data, and making testable predictions for future experiments.

Employment

2024 - present	Assistant Professor , Department of Physics Emory University, Atlanta, GA.
2021- 2024	Member , Simons Center for Systems Biology, School of Natural Sciences Institute for Advanced Study, Princeton, NJ
2017- 2021	Research Associate , Initiative for the Theoretical Sciences The Graduate Center, CUNY, New York, NY
2019	Adjunct Assistant Professor , Department of Physics City College of New York, CUNY, New York, NY
2014-2017	Research Assistant Professor , Simons Center for Geometry and Physics Stony Brook University, Stony Brook, NY

Education

2014	Ph.D. Physics , University of Chicago Thesis: Fractional Quantum Hall Effect in a Curved Space Advisor: Prof. Paul B. Wiegmann
2008	B.A Physics and Applied Mathematics , University of California, Berkeley Honors Thesis: Laser Frequency Stabilization on Cesium Resonance Lines by Doppler Free Magnetic Circular Dichroism Advisor: Prof. Eugene D. Commins

Publications and Preprints

† *senior/corresponding author* ^s *student or postdoc I advised* * *equal contribution*

30. Kanishk Jain^s, Matthew Day^s, **Tankut Can**[†], Emergent retokenization symmetry in large language models: phenomenology and applications, arxiv:2606.15521 (2026),
29. Weishun Zhong, Doron Sivan, **Tankut Can**, Mikhail Katkov, Misha Tsodyks, Semantic Chunking and the Entropy of Natural Language, arXiv:2602.13194 (2026).
28. **Tankut Can**^{*}, Kamesh Krishnamurthy^{*}, Emergence of Robust Memory Manifolds, PRX Life **3**, 023006 (2025). [Featured on Cover](#)
27. **Tankut Can**, Statistical Mechanics of Semantic Compression, arXiv:2503.00612v2 (2025). *Invited submission to Journal of Physics A: Mathematical and Theoretical, Special issue collection: Statistical Physics of Artificial and Biological Neural Networks: Learning, Representation and Computation in Brains and Machines*
26. Weishun Zhong, **Tankut Can**, Antonios Georgiou, Mikhail Katkov, Misha Tsodyks, Random tree model of meaningful memory, Physical Review Letters **134**, 237402 (2025). [Featured in Physics Magazine: How We Remember Stories](#)

25. Antonios Georgiou*, **Tankut Can***, Mikhail Katkov, Misha Tsodyks, Large-scale study of human memory for meaningful narratives, *Learning and Memory* **32**, a054043 (2025).
24. Timothy Doyeon Kim, Thomas Zhihao Luo, **Tankut Can**, Kamesh Krishnamurthy, Jonathan W Pillow, Carlos D Brody, Flow-field inference from neural data using deep recurrent networks, *International Conference on Machine Learning (ICML)*, (2025).
23. Timothy Doyeon Kim, **Tankut Can**[†], Kamesh Krishnamurthy[†], Gated Neural ODEs: Trainability, Expressivity, and Interpretability, *International Conference on Machine Learning (ICML)*, (2023).
22. Aditya Cowsik, **Tankut Can**, Paolo Glorioso, Flatter, faster: scaling momentum for optimal speedup of SGD, *arxiv:2210.16400* (2022)
21. **Tankut Can***, Kamesh Krishnamurthy*, David J. Schwab, Theory of Gating in Recurrent Neural Networks, *Physical Review X* **12**, 011011 (2022)
20. Lucas Sá, Pedro Ribeiro, **Tankut Can**, Tomaž Prosen, Spectral Transitions and Universal Steady-States in Random Kraus Maps and Circuits, *Physical Review B* **102**, 134310 (2020).
19. **Tankut Can***, Kamesh Krishnamurthy*, David J. Schwab, Gating creates slow modes and controls phase-space complexity in GRUs and LSTMs, *Proceedings of The First Mathematical and Scientific Machine Learning Conference, PMLR* **107**, 476-511 (2020).
18. Alexander G. Abanov, **Tankut Can**, Sriram Ganeshan, Gustavo M. Monteiro, Hydrodynamics of two-dimensional compressible fluid with broken parity: variational principle and free surface dynamics in the absence of dissipation, *Physical Review Fluids* **5**, 104802 (2020).
17. **Tankut Can**, Vadim Oganessian, Dror Orgad, and Sarang Gopalakrishnan, Spectral gaps and mid-gap states in random quantum master equations, *Physical Review Letters* **123**, 234103 (2019).
16. **Tankut Can**, Random Lindblad Dynamics, *Journal of Physics A: Mathematical and Theoretical* **52** 485302 (2019).
15. Nathan Schine, Michelle Chalupnik, **Tankut Can**, Andrey Gromov, and Jonathan Simon, Electromagnetic and Gravitational Responses of Photonic Landau Levels, *Nature* **565**, 173 (2019).
14. Alexander G. Abanov, **Tankut Can**, Sriram Ganeshan, Odd surface waves in two-dimensional incompressible fluids, *SciPost Phys.* **5**, 010 (2018).
13. **T. Can**, P. Wiegmann, Quantum Hall states and conformal field theory on a singular surface, *Journal of Physics A: Mathematical and Theoretical* **50**, 494003 (2017). **Invited Paper** for “John Cardy’s scale-invariant journey in low dimensions: a special issue for his 70th birthday”.
12. Dung Xuan Nguyen, **Tankut Can**, and Andrey Gromov, Particle-Hole Duality in the Lowest Landau Level, *Physical Review Letters* **118**, 206602 (2017).
11. **Tankut Can**, Central charge from adiabatic transport of cusp singularities in the quantum Hall effect, *Journal of Physics A: Mathematical and Theoretical* **50**, 174004 (2017). **Invited Paper** for “Emerging Talents” special issue.
10. **T. Can**, Y. H. Chiu, M. Laskin, and P. Wiegmann, Emergent Conformal Symmetry and Geometric Transport Properties of Quantum Hall States on Singular Surfaces, *Physical Review Letters* **117**, 266803 (2016).
9. M. Laskin, **T. Can**, and P. Wiegmann, Collective Field Theory of Quantum Hall States, *Physical Review B* **92**, 235141 (2015).
8. **T. Can**, M. Laskin, P. Wiegmann, Geometry of quantum Hall states: Gravitational anomaly and transport coefficients, *Annals of Physics* **362**, 752 (2015).
7. **T. Can**, P. J. Forrester, G. Téllez, and P. Wiegmann, Exact and Asymptotic Features of the Edge Density Profile for the One Component Plasma in Two Dimensions, *Journal of Statistical Physics* **158**, 1147 (2015).
6. **T. Can**, M. Laskin, and P. Wiegmann, Fractional Quantum Hall Effect in a Curved Space: Gravitational Anomaly and Electromagnetic Response, *Physical Review Letters* **113**, 046803 (2014). **Editors’ Suggestion**
5. **T. Can**, P. J. Forrester, G. Téllez, and P. Wiegmann, Singular Behavior at the Edge of Laughlin States, *Physical Review B* **89**, 235137 (2014).
4. Kenley M. Pelzer, **Tankut Can**, Stephen K. Gray, Dirk K. Morr, and Gregory S. Engel, Coherent Transport and Energy Flow Patterns in Photosynthesis under Incoherent Excitation, *Journal of Physical Chemistry B* **118**(10), 2693 (2014).
3. Joel Mabillard, **Tankut Can**, and Dirk K. Morr, Spatial current patterns, dephasing and current imaging in graphene nanoribbons, *New Journal of Physics* **16**, 013054 (2014).
2. **Tankut Can** and Dirk K. Morr, Atomic Resolution Imaging of Currents in Nanoscopic Quantum Networks via Scanning Tunneling Microscopy, *Physical Review Letters* **110**, 086802 (2013).
1. **Tankut Can**, Hui Dai, and Dirk K. Morr, Current eigenmodes and dephasing in nanoscopic quantum networks, *Physical Review B* **85**, 195459 (2012).

Research Group

- 2024 – present **Kanishk Jain**, Postdoc
- Retokenization symmetry in large language models
- 2025 – present **Matthew Day**, PhD student, Physics
- Physics of multi-agent systems and retokenization symmetry
- 2024 – present **Roberto Avalos**, PhD student, Physics
- Evolution of language statistics under renormalization group flow
- 2025 – present **Eden Zhu**, PhD student, Neuroscience
- Human narrative memory and retrieval

Honors and Awards

- 2026 Invited paper for “Statistical Physics of Artificial and Biological Neural Networks” special issue of J. Phys. A: Math. Theor.
- 2025 PRX Life [cover](#), paper “Emergence of Robust Memory Manifolds”
- 2025 PRL featured in Physics Magazine: [How We Remember Stories](#)
- 2018 General workshop award (\$20,000) from Institute for Complex Adaptive Matter (ICAM) for “Machine Learning and Statistical Physics” (with S. Gopalakrishnan, V. Oganessian, and D. Schwab)
- 2016 Invited paper for “Emerging Talents” special issue of J. Phys. A: Math. Theor.
- 2013 Gregor Wentzel Research Prize, University of Chicago, for outstanding work in theoretical physics
- 2010-2012 University of Chicago Robert A. Millikan Fellowship (U.S. Department of Education Graduate Assistance in Areas of National Need (GAANN) Fellowship)
- 2008 Distinction in general scholarship, University of California, Berkeley
- 2008 High Honors in Physics, University of California, Berkeley
- 2006-2007 Isidore Pomerantz Scholarship in Physics, University of California, Berkeley

Invited Talks

27. “Memory for Narratives and the Entropy of English”, Fundamental principles of Learning and Representation: from Brains to LLMs, EPFL, Lausanne, Switzerland, May 12, 2026.
26. “The Structure of Memory for Narratives”, AI4Science Seminar, Georgia Institute of Technology, Atlanta, GA, March 31, 2026
25. “The Structure of Memory for Narratives”, Computational Cognition (CoCo) Conference, Georgia Institute of Technology, Atlanta, GA, March 13, 2026
24. “The Structure of Memory for Narratives”, INNSight Forum Seminar, Institute for Neuroscience, Neurotechnology, and Society (INNS), Georgia Institute of Technology, Atlanta, GA, Nov. 10, 2025
23. “Two views of semantic compression”, Physics of Learning, Johns Hopkins University, Washington, D.C., April 16, 2025
22. “Large-scale study of human memory for meaningful narratives”, Symposium on Learning, with and without neurons, CUNY Graduate Center, New York, NY, Nov. 22, 2024.
21. “Statistical Mechanics of Semantic Compression”, American Physical Society March Meeting, Minneapolis, MN, 2024
20. “Chatbots for Science”, CCNY Physics Colloquium, Nov. 15, 2023.
19. “Physics of Computation in Brains and Machines”, Emory Physics Colloquium, Feb. 23, 2023.
18. “Chaos and Computation in Gated Neural Networks”, Department of Neurobiology and Behavior, Stony Brook University, Feb. 14, 2023
17. “Dynamics and Computation in Gated Recurrent Neural Networks”, Frontiers in Applied and Computational Mathematics, May 21, 2022.
16. “Physics of Computation in Brains and Machines”, Cooper Union Special Seminar, April 28, 2022.
15. “Dynamics and Computation in Gated Recurrent Neural Networks”, Brown Theoretical Physics Center, Jan. 21, 2022.

14. "Random Dissipative Quantum Evolution", Many Body Physics in Open Quantum Systems, Princeton Center for Theoretical Physics, Princeton, New Jersey, Jan. 28, 2021.
13. "Dynamics of Gated Recurrent Neural Networks", Simons Center for Systems Biology, Institute for Advanced Study, Princeton, New Jersey, Jan. 8, 2021.
12. "Dynamics of Gated Recurrent Neural Networks", City College of New York Physics Virtual Colloquium, September 16, 2020.
11. "Spectral Gaps and Mid-gap modes in Random Quantum Markov Master Equations", Universality and Ergodicity Program Seminar, Simons Center for Geometry and Physics, Stony Brook, NY, October 3, 2019.
10. "Coulomb Gas in Superspace", New Directions in Mathematics of Coulomb Gases and Quantum Hall Effect, Mittag-Leffler Institute, Djursholm, Sweden, July 4, 2019.
9. "Geometric Response of Quantum Hall Liquids", High Energy Physics seminar, City College of New York, New York, March 22, 2019.
8. "Spectral Gaps in Random Master Equations", Snapshots of quantum dynamics: constraints, integrability and hidden orders, Graduate Center, City University of New York, New York, March 11, 2019.
7. "Adiabatic Charge and Momentum Transport in the Fractional Quantum Hall Effect", Kadanoff Center for Theoretical Physics weekly seminar, Chicago, IL, Nov. 6, 2017.
6. "Probing Quantum Hall States with Singularities and Defects" Symposium on new results in collective phenomena, Graduate Center, City University of New York, New York, January 31, 2017.
5. "Probing Quantum Hall States with Flux Tubes, Cones, and Cusps", Geometrical Degrees of Freedom in Topological Phases, Banff Institute for Research Sciences, Banff, Canada, August 24, 2016.
4. "Probing Quantum Hall States with Flux Tubes and Cones", Workshop on Geometric Aspects of the Quantum Hall Effect, University of Cologne, Cologne, Germany, December 15, 2015.
3. "Collective Field Theory of Quantum Hall states as Random Geometry", Quantum Geometry, Stochastic Geometry, Random Geometry, you name it, Simons Center for Geometry and Physics, Stony Brook, NY, June 17, 2015.
2. "Ward Identities for Fractional Quantum Hall states", Program: Large N limit problems in Kahler geometry, Simons Center for Geometry and Physics, Stony Brook, NY, May 28, 2015
1. "Singular behavior at the edge of fractional quantum Hall states", Kadanoff Center for Theoretical Physics weekly seminar, Chicago, IL, Oct. 14, 2013.

Peer-Reviewed Conference Abstracts

7. K. Jain^s, M. Day^s, T. Can, "Generating output diversity from prompt retokenization", Workshop on Scientific Methods for Understanding Deep Learning, International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, April 2026.
6. W. Zhong, T. Can, M. Katkov, M. Tsodyks, "Semantic Chunking and the Entropy of Natural Language", Computational and Systems Neuroscience (CoSyNe), Lisbon, PT, March 2026.
5. W. Zhong, T. Can, A. Georgiou, I. Shnayderman, M. Katkov, M. Tsodyks, "A Statistical Theory of Sequence Compression in Human Memory", Computational and Systems Neuroscience (CoSyNe), Montreal, CA, March 2025.
4. A. Georgiou, T. Can, M. Katkov, M. Tsodyks, "Large Scale Study of Human Memory for Narratives using Large Language Models", Computational and Systems Neuroscience (CoSyNe), Lisbon, PT, March 2024.
3. Timothy Doyeon Kim, Tankut Can, Kamesh Krishnamurthy, "Learning and Shaping Manifold Attractors for Computation in Gated Neural ODEs", accepted to Symmetry and Geometry in Neural Representations (NeurReps) Workshop at the 36th Conference on Neural Information Processing Systems (NeurIPS) 2022.
2. T. Can, K. Krishnamurthy, D. J. Schwab, "Dynamics of continuous-time gated recurrent neural networks", accepted to Machine Learning and the Physical Sciences Workshop at the 34th Conference on Neural Information Processing Systems (NeurIPS) 2020.
1. K. Krishnamurthy, T. Can, D. J. Schwab, "Theory of gating in recurrent neural networks", Computational and Systems Neuroscience (CoSyNe), Denver, CO, USA, February 2020.

Contributed Talks and Presentations

31. "Semantic Chunking and the Entropy of Natural Language", Authors: W. Zhong, T. Can, M. Katkov, M. Tsodyks, Global Physics Summit, Denver, CO, 2026
30. "Tokenization gauge symmetry in language modeling", Authors: K. Jain^s, M. Day^s, T. Can[†], Global Physics Summit, Denver, CO, 2026
29. "Evolution of Language Statistics under Renormalization Group Flow", Authors: R. Avalos^s, T. Can[†], Global Physics Summit, Denver, CO, 2026
28. "Aspects of Tokenization in Language Modeling", Theoretical Physics for Artificial Intelligence, Aspen Center for Physics, January 11, 2026
27. "Dynamics of Neural Networks with Nonlinear Synaptic Interactions", American Physical Society March Meeting, Anaheim, CA, 2025
26. "LLM-assisted study of human memory for meaningful narratives", KITP program "Deep Learning from the Perspective of Physics and Neuroscience", Nov. 21, 2023
25. "Robust Memory Manifolds in Neural Networks", American Physical Society March Meeting, Las Vegas, NV, 2023.
24. "Emergence of Memory Manifolds", Contributed Talk, From Neuroscience to Artificially Intelligent Systems (NAISys), April 6, 2022.
23. "Gated recurrent neural networks 2: a novel first-order chaotic transition", American Physical Society March Meeting, virtual meeting, 2021.
22. "Theory of Gating in Recurrent Neural Networks", co-presentation with K. Krishnamurthy, Facebook AI research, Nov. 13, 2020.
21. "Gating Creates Slow Modes and Controls Phase-space Complexity in GRUs and LSTMs", Mathematical and Scientific Machine Learning Virtual Conference July 20, 2020.
20. "Dynamics of Gated Recurrent Neural Networks", Initiative for the Theoretical Sciences special seminar, CUNY Graduate Center, New York, NY, Feb. 19, 2020.
19. Journal Club: "Dynamics of stochastic gradient descent for two-layer neural networks in the teacher-student setup", Center for the Physics of Biological Function, Princeton University, Princeton, NJ, Feb. 13, 2020.
18. "Random Lindblad Dynamics", American Physical Society March Meeting, Boston, MA, 2019.
17. Journal Club: "Path Integral for Random Neural Networks", Initiative for the Theoretical Sciences Journal Club, CUNY Graduate Center, New York, NY, Oct. 15 & 23, 2018.
16. "Random Lindblad Dynamics", Random Matrices, Integrability and Complex Systems, Yad Hashmona, Israel, Oct. 4, 2018.
15. "Non-dissipative odd viscosity in quantum and classical systems", Statistical techniques for correlation analysis: Quantum Many-body Systems and more, Centro Internacional de Ciencias A.C., Cuernavaca, México, July 18, 2018.
14. Journal Club: "Eigenvector Statistics of Random Non-Hermitian Matrices", Initiative for the Theoretical Sciences Journal Club, CUNY Graduate Center, New York, NY, June 4, 2018.
13. "Edge Waves in Odd Fluids", American Physical Society March Meeting, Los Angeles, CA, March 2018.
12. "Coulomb Plasma on a Singular Surface", Summer School: Dyson-Schwinger equations, topological expansions, and random matrices, Columbia University, New York, NY, Aug. 28, 2017.
11. "Adiabatic Charge and Momentum Transport in the Fractional Quantum Hall Effect", Mathematics of Topological Phases of Matter, Simons Center for Geometry and Physics, Stony Brook, NY, June 15, 2017.
10. "Central Charge from Adiabatic Transport of Cusp Singularities in the Quantum Hall Effect", American Physical Society March Meeting, New Orleans, LA, March 2017.
9. "Beyond the Plasma Analogy: Collective Field Theory for Quantum Hall States", American Physical Society March Meeting, Baltimore, MD, 2016.
8. "Introduction to the Quantum Hall Effect and Geometry", Simons Center for Geometry and Physics weekly seminar, Feb. 23, 2015.
7. "Fractional Quantum Hall Effect in a Curved Space", Non-Hermitian Random Matrices: 50 years after Ginibre, Research Workshop of the Israel Science Foundation, Yad Hashmona, Israel, Oct. 26, 2014.
6. "Fractional Quantum Hall Effect in a curved space", Condensed Matter Theory seminar, Stony Brook University, Sept. 22 and 29, 2014.

5. Journal Club: “Laughlin wave function and anyons”, Kadanoff Center for Theoretical Physics, University of Chicago, Chicago, IL, May 2, 2014.
4. Journal Club: “Non-equilibrium dynamics of entanglement entropy quench”, Kadanoff Center for Theoretical Physics, University of Chicago, Chicago, IL, Feb. 27, 2014
3. Journal Club: “3D Topological Insulators”, Kadanoff Center for Theoretical Physics, University of Chicago, Chicago, IL, Nov. 7, 2013
2. “Imaging Spatial Current Eigenmodes in Nanoscopic Quantum Networks”, American Physical Society March Meeting, Boston, MA, 2012
1. “Imaging Spatial Current Eigenmodes in 2D Nanostructures”, Nano Talk Student Symposium, University of Chicago, Chicago, IL, 2011.

Posters

9. Eden Zhu^s, Tankut Can, “Human episodic memory and retrieval in humans”, abstract and poster at ACSFN (Atlanta chapter, Society of Neuroscience), Emory University, 2025.
8. Eden Zhu^s, Tankut Can, “Human episodic memory and retrieval in humans”, poster at Emory Neuroscience Graduate Program event, Emory University, 2025.
7. A. Georgiou, T. Can, M. Katkov, M. Tsodyks, “Large Scale Study of Human Memory for Narratives using Large Language Models”, Context and Episodic Memory Symposium, Philadelphia, PA, May 2024.
6. Aditya Cowsik, Tankut Can, Paolo Glorioso, “Flatter, Faster: Scaling Momentum for Optimal Speedup of SGD”, ICML Workshop on High-Dimensional Learning, 2023
5. Aditya Cowsik, Tankut Can, Paolo Glorioso, “Momentum matters: Acceleration and generalization with SGD and label noise”, Conference on the Mathematical Theory of Deep Neural Networks (Deep Math), San Diego, CA, Nov. 17-18, 2022.
4. T. Can, K. Krishnamurthy, D. J. Schwab, “Gating creates slow modes and control phase space complexity in GRUs and LSTMs”, 14th Annual Machine Learning Symposium, New York Academy of Sciences, New York, NY, March 13, 2020 (took place virtually).
3. T. Can, Adiabatic Charge and Momentum Transport in the Fractional Quantum Hall Effect, Summer School: Dyson-Schwinger equations, topological expansions, and random matrices, Columbia University, New York, NY, Aug. 28, 2017.
2. T. Can, Adiabatic Charge and Momentum Transport in the Fractional Quantum Hall Effect, Frontiers in Emergent Quantum Phenomena, New York University, New York, NY, 2017.
1. T. Can, H. Dai, D. K. Morr, Imaging Real Space Currents in Nanostructures, Electronic Transport in Nanoengineered Materials Workshop, University of Chicago, Chicago, IL, 2010.

Service and Outreach

Peer Review

Referee: Physical Review Letters, Physical Review B, Physical Review E, Annals of Physics, Journal of Physics A: Mathematical and Theoretical, Journal of Physics Communications, Trends in Hearing.

2025	Reviewer, New Frontiers in Associative Memories workshop, ICLR
2023	Program Committee, Associative Memory and Hopfield Networks workshop, NeurIPS

Conference and Workshop Organization

2018 - 2020	Organizer, Machine Learning and Statistical Physics journal club, Initiative for the Theoretical Sciences, CUNY
2018	Organizer (with S. Gopalakrishnan, V. Oganessian, and D. Schwab), Workshop: Machine Learning and Statistical Physics, The Graduate Center, CUNY, New York, NY, Nov. 13–15
2016	Organizer (with A. Abanov, A. Kapustin, and P. Wiegmann), Program: Geometry of Quantum Hall States, Simons Center for Geometry and Physics, Stony Brook University, April 18 – June 17
2016	Organizer (with A. Abanov, A. Kapustin, and P. Wiegmann), Workshop: Geometry of Quantum States in Condensed Matter Systems, Simons Center for Geometry and Physics, Stony Brook University, April 18–22
2016	Session Chair, APS March Meeting, Session: Fractional QHE: Level Mixing & Transitions

Departmental Service

2024 - 2025 Graduate Admissions Committee; Colloquium Committee, Department of Physics, Emory University

Outreach and Training

- 2025 Participant, workshop Studying AI at Scale, Perimeter Institute. A working group for building a diverse community of scientists in the science of AI
- 2024 Public Lecture “Nobel Prize: Physics and AI”, Department of Physics, Emory University
- 2023 Contribution to [Campus Conversations](#) on Artificial Intelligence, Institute for Advanced Study, August 17
- 2023 Public talk “Exploring Memory with AI”, “Dinner with a Member” series, Institute for Advanced Study, April 26
- 2021 Tutor, Abdus Salam International Centre for Theoretical Physics Virtual School “The Hitchhiker’s Guide to Condensed Matter and Statistical Physics: Machine Learning and Condensed Matter”, on machine learning and neural networks, January 13
- 2020 Tutor, Physics of Life Summer School, Center for the Physics of Biological Function, Princeton University, on “Statistical Mechanics of Machine Learning”, August 3

Member, American Physical Society

Teaching

- 2026 Thermodynamics and Statistical Physics (Physics 421), Emory University, Fall Semester
 - Lecture-only course with active learning in-class problem solving component. [Course website](#).
- 2026 Emergent and Collective Phenomena (Physics 504), Emory University, Spring Semester
 - Graduate level lecture-only course on statistical mechanics [Course website](#).
- 2025 Thermodynamics and Statistical Physics (Physics 421), Emory University, Fall Semester
 - Lecture-only course with active learning in-class problem solving component. [Course website](#).
- 2024 Physics of AI, Special topics in theoretical physics (Physics 731R), Emory University, Fall Semester
 - Organized a seminar-style course on modern machine learning and AI, with special focus on theoretical physics-inspired approaches to understanding AI. More details, including reading list, can be found on the archival [course website](#).
- 2019 General Physics I (Physics 203), City College of New York, CUNY, Summer Session
 - Designed and taught introductory undergraduate physics course for life sciences majors. More details, including my original exam questions, can be found on the archival [course website](#).
- 2017 Instructor, Stony Brook University, Stony Brook, NY, Classical Physics II (Physics 132), Spring Semester
 - Weekly duties included teaching 1 hour recitation section, and grading homework.
- 2008-2012 Graduate Teaching Assistant, University of Chicago, Undergraduate physics courses for majors, 8 semesters total
 - Weekly duties included teaching 1 hour discussion section, 4 hour lab sections, 2 hours of office hours, and grading homework, labs, and exams.
- 2007 Undergraduate Student Instructor, University of California, Berkeley, Physics for Scientists and Engineers (Physics 7B), Fall Semester
 - Duties included 4 hours laboratory/discussion per week, grading homework, labs and exams.